

1. Creating and Renaming Files/Directories

```
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ deactivate
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ mkdir test_dir
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ cd test_dir
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ touch example.txt
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ ls
example.txt
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ mv example.txt renamed_example.txt
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ ls
renamed_example.txt
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ |
```

`mkdir test_dir` creates a new directory named `test_dir`

`cd test_dir` changes the current directory to `test_dir`

`touch example.txt` creates an empty file named `example.txt`

`mv example.txt renamed_example.txt` renames file from `example.txt` to `renamed_example.txt`

`ls` lists the files in the directory to verify that the file has been renamed successfully

2. Viewing File Contents

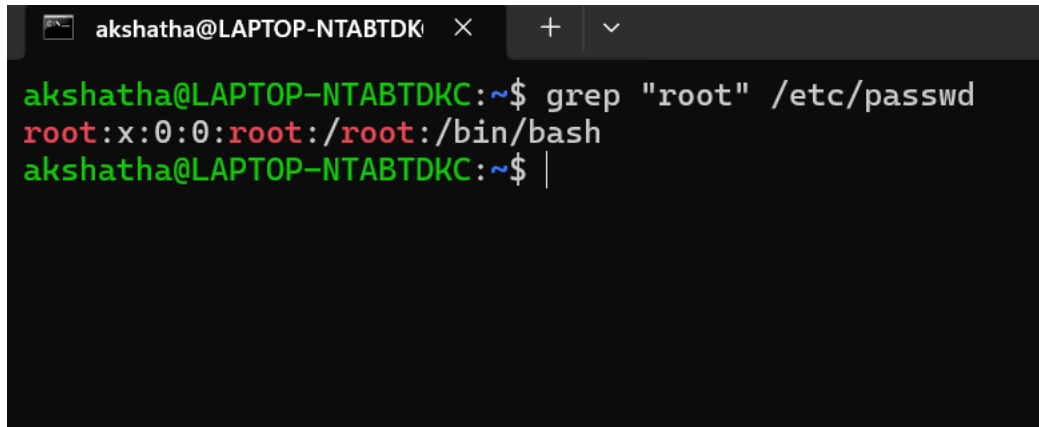
```
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uidd:x:103:103::/run/uidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
akshatha:x:1000:1000:::/home/akshatha:/bin/bash
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ head -5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ tail -5 /etc/passwd
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uidd:x:103:103::/run/uidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
akshatha:x:1000:1000:::/home/akshatha:/bin/bash
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial/test_dir$ |
```

cat /etc/passwd displays the complete contents of the /etc/passwd file

head -n 5 /etc/passwd displays the first 5 lines of the file

tail -n 5 /etc/passwd displays the last 5 lines of the file

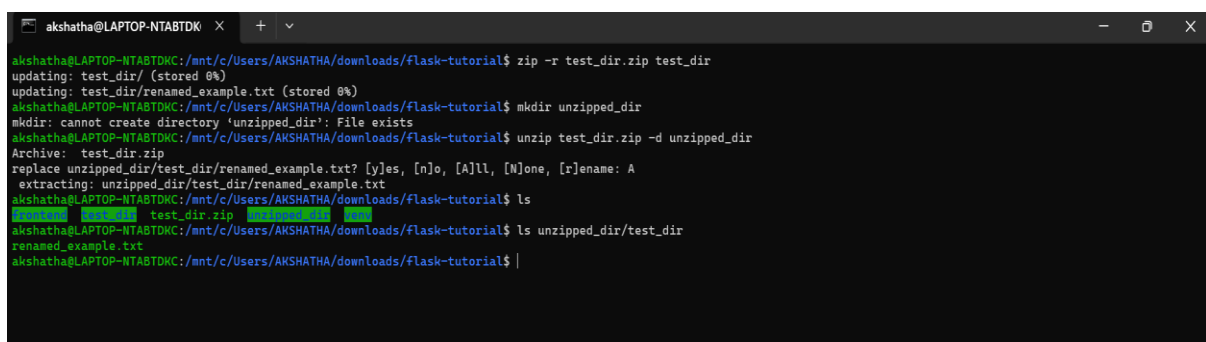
3. Searching for Patterns

A terminal window titled 'akshatha@LAPTOP-NTABTDK' shows a command prompt where the user enters 'grep "root" /etc/passwd'. The output shows a single line: 'root:x:0:0:root:/root:/bin/bash'. The prompt then returns to the user's shell.

```
akshatha@LAPTOP-NTABTDK:~$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
akshatha@LAPTOP-NTABTDK:~$ |
```

grep “root” /etc/passwd searches the /etc/passwd file for all lines containing the word “root”

4. Zipping and Unzipping

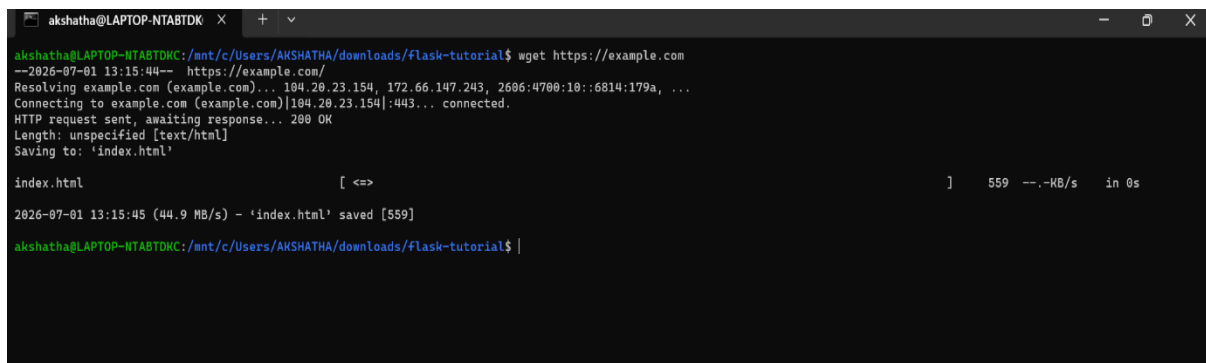
A terminal window titled 'akshatha@LAPTOP-NTABTDK' shows a series of commands. First, 'zip -r test_dir.zip test_dir' is run, creating a ZIP file. Then, 'mkdir unzipped_dir' is run, but it fails because the directory already exists. Next, 'unzip test_dir.zip -d unzipped_dir' is run, successfully extracting the contents. Finally, 'ls unzipped_dir/test_dir' is run, showing the files 'renamed_example.txt' and 'test_dir.zip'.

```
akshatha@LAPTOP-NTABTDK:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ zip -r test_dir.zip test_dir
updating: test_dir/ (stored 0%)
updating: test_dir/renamed_example.txt (stored 0%)
akshatha@LAPTOP-NTABTDK:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ mkdir unzipped_dir
mkdir: cannot create directory 'unzipped_dir': File exists
akshatha@LAPTOP-NTABTDK:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ unzip test_dir.zip -d unzipped_dir
Archive:  test_dir.zip
replace unzipped_dir/test_dir/renamed_example.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
extracting: unzipped_dir/test_dir/renamed_example.txt
akshatha@LAPTOP-NTABTDK:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ ls
test_dir.zip  unzipped_dir
akshatha@LAPTOP-NTABTDK:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ ls unzipped_dir/test_dir
renamed_example.txt
akshatha@LAPTOP-NTABTDK:/mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ |
```

zip -r test_dir.zip test_dir compresses the test_dir directory and all its contents into a ZIP file named test_dir.zip The -r option means recursive, so it includes all files and subdirectories

unzip test_dir.zip -d unzipped_dir extracts the contents of the ZIP file into a new directory named unzipped_dir

5. Downloading Files



```
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ wget https://example.com
--2026-07-01 13:15:44-- https://example.com/
Resolving example.com (example.com)... 104.20.23.154, 172.66.147.243, 2606:4700:10::6814:179a, ...
Connecting to example.com (example.com)|104.20.23.154|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'

index.html                                [ <=> ] 559 --KB/s in 0s

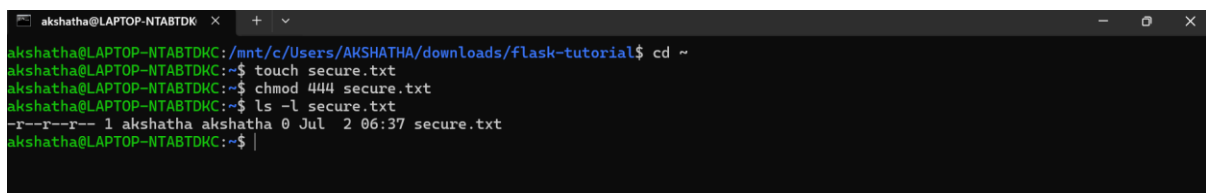
2026-07-01 13:15:45 (44.9 MB/s) - 'index.html' saved [559]

akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ |
```

wget is a command-line utility used to download files from the internet

In this task, `wget https://example.com` downloads the homepage of the website `example.com` and saves it as `index.html` in the current directory

6. Changing Permissions



```
akshatha@LAPTOP-NTABTDK: /mnt/c/Users/AKSHATHA/downloads/flask-tutorial$ cd ~
akshatha@LAPTOP-NTABTDK: ~$ touch secure.txt
akshatha@LAPTOP-NTABTDK: ~$ chmod 444 secure.txt
akshatha@LAPTOP-NTABTDK: ~$ ls -l secure.txt
-r--r--r-- 1 akshatha akshatha 0 Jul 2 06:37 secure.txt
akshatha@LAPTOP-NTABTDK: ~$ |
```

`touch secure.txt` creates an empty file named `secure.txt`.

`chmod 444 secure.txt` changes the file permissions to read-only for the owner, group, and others.

`ls -l secure.txt` displays the file permissions to verify that the changes were applied successfully.

7. Working with Environment Variables



```
akshatha@LAPTOP-NTABTDK: ~$ export MY_VAR="Hello, Linux!"
akshatha@LAPTOP-NTABTDK: ~$ echo $MY_VAR
Hello, Linux!
akshatha@LAPTOP-NTABTDK: ~$ |
```

`export MY_VAR="Hello, Linux!"` creates an environment variable named `MY_VAR` and assigns it the value `"Hello, Linux!"`.

`echo $MY_VAR` displays the value of the environment variable.